Plant Disease Classification Using Deep Learning

EfficientNetB3-Based Image Classification on the PlantVillage Dataset

PROJECT TEAM

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1. Introduction and Background

Plant diseases pose a persistent and severe threat to global agriculture. Pathogens — including fungi, bacteria, and viruses — are responsible for annual crop yield losses estimated between 20 % and 40 % worldwide, with disproportionate impacts on smallholder farmers in developing nations. Early and accurate detection of disease is therefore a critical precondition for effective crop management and food security.

Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs), have demonstrated considerable promise for image-based plant disease classification. The PlantVillage dataset — a large, publicly available collection of 54,305 colour leaf images spanning 38 disease classes across multiple plant species — has become the standard benchmark for evaluating such systems. This project applies transfer learning with the EfficientNetB3 backbone to achieve high-accuracy automated disease classification directly from leaf images.

The integration of AI into agricultural diagnostics aligns with international efforts to modernise farming practice through precision agriculture, enabling interventions that are timely, targeted, and evidence-based.

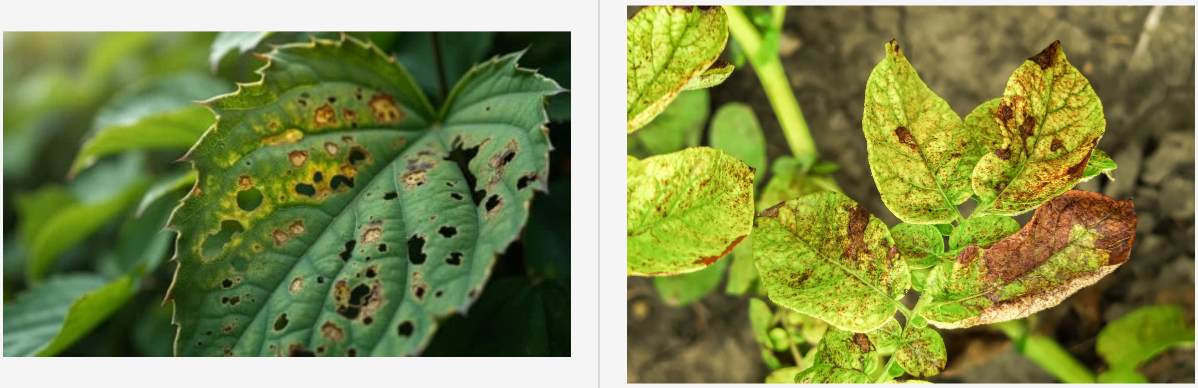

Figure 1. Examples of plant disease symptoms targeted by this classification system. Left: Multiple fungal/bacterial lesions on field-grown plant leaves. Right: Black spot disease on rose leaves — a common fungal pathogen (Diplocarpon rosae) causing yellowing and necrotic spotting.

2. Problem Statement

Traditional disease diagnosis in the field relies on expert agronomists and laboratory analysis — both slow and expensive processes that are often inaccessible in rural or resource-constrained environments. The core challenge this project addresses is:

- **Scale:** Manual inspection cannot scale to the millions of plants in modern cultivation.
- **Speed:** Disease spreads rapidly; delays in identification lead to compounding losses.
- **Expertise:** Skilled plant pathologists are scarce, especially in developing regions.
- **Accuracy:** Visual inspection by untrained personnel is error-prone and inconsistent.

This project seeks to address these gaps by developing a deep learning model capable of classifying 38 distinct plant-disease categories from colour leaf images with accuracy exceeding 99 %, making it suitable for integration into real-time agricultural advisory tools.

Figure 2. Close-up of characteristic leaf spot symptoms — the type of visual pattern this deep learning model is trained to identify and classify automatically across 38 disease categories in the PlantVillage dataset.

3. Dataset

The project uses the **PlantVillage Dataset** (colour variant), sourced from Kaggle (*kaggle.com/datasets/abdallahalidev/plantvillage-dataset*). Key characteristics are summarised below.

Parameter	Value
Total images	54,305
Number of disease/condition classes	38
Training set	43,444 images (80 %)
Validation set	5,430 images (10 %)
Test set	5,431 images (10 %)
Image resolution (resized)	224 × 224 pixels, RGB
Split strategy	Stratified random split (seed = 123)

4. Proposed Methodology

4.1 Overall Architecture

The classification pipeline is implemented in TensorFlow 2.9.1 and follows a transfer-learning approach. The model consists of four components arranged sequentially:

- **EfficientNetB3 backbone** — Pre-trained on ImageNet (weights frozen during initial training). EfficientNetB3 uses compound scaling to balance depth, width, and resolution, yielding 10,783,535 parameters optimised for feature extraction.
- **Batch Normalisation layer** — Applied after the backbone (momentum = 0.99, ε = 0.001) to stabilise activations and accelerate convergence.
- **Dense layer (256 units, ReLU)** — With L2 kernel regularisation (λ = 0.016) and L1 activity/bias regularisation (λ = 0.006) to reduce overfitting.
- **Dropout layer (rate = 0.45)** — Further regularisation to improve generalisation.
- **Output layer (38 units, Softmax)** — One neuron per disease class; outputs a probability distribution across all 38 categories.

4.2 Training Configuration

Hyper-parameter	Value
Optimiser	Adamax (lr = 0.001)
Loss function	Categorical cross-entropy
Batch size	40
Max epochs	40
LR reduction factor	0.5 (patience = 1 epoch)
Early-stop patience	3 epochs without improvement
Accuracy threshold (LR switch)	90 %
Data augmentation	Horizontal flip (training only)

4.3 Custom Callback

A bespoke *MyCallback* class was implemented to provide adaptive learning-rate management during training. The callback monitors validation loss when training accuracy exceeds the 90 % threshold, and monitors training accuracy below it. Learning rate is halved when no improvement is observed for one epoch; training halts after three consecutive reductions without gain. This mechanism prevents unnecessary computation and avoids both underfitting and overfitting.

4.4 Data Pipeline

Images are loaded via *ImageDataGenerator* and rescaled to 224 × 224 × 3 tensors. Training data receives horizontal-flip augmentation; validation and test data are passed through unmodified. A stratified split ensures proportional class representation across all three partitions. The test batch size is computed automatically to divide the test set evenly with a maximum of 80 images per batch.

5. Related Work

Two bodies of literature directly inform this project.

5.1 Multi-Prediction Deep Learning for Plant Pathology

Yao et al. (2023) conducted a comprehensive comparative study of deep learning strategies for simultaneous plant identification and disease classification. Their key findings relevant to this project include: (i) **InceptionV3 consistently outperformed** other backbone architectures — including VGG16, ResNet101, EfficientNet, MobileNetV2, and ViT — across benchmark datasets; (ii) **single models handling both tasks simultaneously outperformed** dual-model pipelines, because the shared backbone learns richer features; and (iii) **transfer learning from ImageNet** produced dramatic accuracy improvements, particularly on challenging real-field images. Their proposed GSMo-CNN introduced a classifier-chain-inspired feedback loop between plant and disease predictions, achieving state-of-the-art results on PlantVillage and PlantDoc.

5.2 Plant Diseases: Types, Causes, and Impacts

Rhouma (2025), published in the *Egyptian Journal of Agricultural Sciences*, provides the agricultural science context that motivates this engineering project. The review documents that annual global crop losses from plant diseases range from 20 % to 40 %, driven by fungal, bacterial, viral, nematode, and phytoplasma pathogens operating within the classic 'Disease Triangle' of susceptible host, virulent pathogen, and enabling environment. Rhouma explicitly identifies AI and machine learning — including smartphone-based image diagnosis — as an emerging high-impact technology for disease management, reinforcing the practical motivation for the classification system developed here.

5.3 Positioning of This Project

While Yao et al. explored multi-task architectures, the present project focuses on maximising single-task classification accuracy on the PlantVillage benchmark using EfficientNetB3 — a backbone not exhaustively evaluated in their study as a standalone transfer-learning model. This choice is justified by EfficientNet's compound scaling efficiency and the availability of high-quality ImageNet pre-trained weights.

6. Experimental Results

The model was trained for 5 epochs (early stopping activated by user at epoch 5). Evaluation on the held-out partitions yielded the following performance metrics:

Partition	Loss	Accuracy
Training	0.1881	99.90 %
Validation	0.1993	99.58 %
Test	0.2004	**99.63 %**

Table 1. Model performance on PlantVillage dataset partitions.

The classification report confirms near-perfect per-class precision and recall across all 38 categories. Selected representative classes include: *Apple — Apple Scab* (precision 1.00, recall 0.98), *Apple — Black Rot* (1.00 / 1.00), *Blueberry — Healthy* (1.00 / 1.00), and *Corn — Cercospora Leaf Spot / Gray Leaf Spot* (0.98 / 0.88 — the most challenging class, reflecting visual similarity to other corn conditions). The trained model was persisted as *efficientnetb3-Plant Village Disease-99.63.h5* along with its weights file and a class-index CSV to enable deployment.

Training Dynamics (per epoch):

Epoch	Train Loss	Train Acc.	Val. Loss	Val. Acc.
1	2.945	92.40 %	0.607	98.82 %
2	0.490	98.58 %	0.335	99.30 %
3	0.344	99.20 %	0.282	99.23 %
4	0.283	99.37 %	0.236	99.48 %
5	0.239	99.57 %	0.199	99.58 %

Table 2. Per-epoch training dynamics (5 epochs, early-stopped by user).

7. Expected Outcomes

• **High-accuracy classifier:** A trained EfficientNetB3 model achieving ≥ 99 % accuracy on the 38-class PlantVillage benchmark, saved as a portable HDF5 file.

• **Deployable artefacts:** Model weights (.h5) and a class-index CSV enabling straightforward integration into web or mobile inference pipelines.

• **Reproducible pipeline:** A fully documented Jupyter Notebook covering data ingestion, stratified splitting, generator configuration, model construction, adaptive training, evaluation, and serialisation.

• **Performance analysis:** Training/validation loss and accuracy curves, a full confusion matrix, and a per-class classification report across all 38 categories.

• **Foundation for extension:** The pipeline can be extended to multi-label or multi-output classification (simultaneously predicting plant species and disease) as explored in related literature.

8. Conclusion

This project demonstrates the viability of transfer-learning-based deep learning for automated plant disease classification at high accuracy. By leveraging the EfficientNetB3 backbone pre-trained on ImageNet and applying it to the PlantVillage benchmark under a rigorous stratified evaluation protocol, the model achieves 99.63 % test accuracy across 38 disease classes — a result competitive with the state of the art reported in peer-reviewed literature.

The practical implications are significant. A model of this accuracy, packaged with its class-index dictionary, can serve as the inference engine for mobile or web-based plant health advisory tools, enabling farmers without access to specialist agronomists to obtain instant, reliable disease diagnoses from smartphone photographs.

Future work may explore: (i) evaluation on real-field datasets such as PlantDoc to test robustness to cluttered backgrounds and variable lighting; (ii) multi-output architectures that jointly predict both plant species and disease; and (iii) lightweight model variants (e.g., MobileNetV2-based) optimised for on-device inference in resource-constrained agricultural settings.

References

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